Daixin Tian

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EDUCATION

MSc in Applied Computing, Artificial Intelligence Concentration September 2025 – December 2026 (expected)
University of Toronto, Department of Computer Science
Relevant Courses: Neural Network and Deep Learning, Digital Image Processing, Topics in Software Engineering

Bachelor of Applied Science and Engineering September 2020 – June 2025
University of Toronto, Department of Electrical and Computer Engineering
Major: Computer Engineering Minor: Artificial Intelligence

WORK EXPERIENCE

Machine Learning Research Engineer May 2026 – Present
Huawei Noah's Ark Lab | Internship *Markham, ON, Canada*
- Developed a fully automated Unreal Engine 5 MetaHuman pipeline for scalable 3D video data generation, filling a gap in existing tools by producing photorealistic multi-character videos with high-fidelity facial animation and diverse camera trajectories.
- Researching and benchmarking evaluation protocols for 3D camera-controlled video editing, focusing on camera trajectory accuracy, temporal consistency, and identity preservation.
- Training and optimizing large-scale video diffusion models with PyTorch on multi-node environments.

AI Research Assistant January 2025 – Present
University of Toronto | Work Study Program *Toronto, ON, Canada*
- Developed a **Retrieval-Augmented Generation (RAG)** chatbot using Large Language Models (LLMs) for the Toronto and Region Conservation Authority.
- Integrated Streamlit web interface with multi-session chat, API key configuration, and document-aware conversation history.
- Designed and executed evaluation scripts for prompt engineering using BLEU, ROUGE metrics, and Ragas to assess performance across prompts, improving answer relevance by ~**22%** after iterative refinements.

Machine Learning Research Engineer May 2023 – May 2024
Huawei Noah's Ark Lab | Internship *Markham, ON, Canada*
- Researched and developed Text-to-Image and Text-to-Video generative models, including **diffusion models and transformers**, and implemented optimization techniques for training and inference from CVPR and AAAI papers.
- Contributed to bug fixing and optimization of the training code of PixArt-alpha, a **3.2k+ star GitHub** repository for text-to-image generative modeling: github.com/orgs/PixArt-alpha/discussions/119.
- Developed and optimized algorithms that reduced false triggering of a smartphone voice assistant by **80%**.
- Designed and tested a generative model for poster layout generation based on text and image content.

OTHER EXPERIENCE

Independent Developer, AI Interview Training Coach, Higgs Audio v2 Hackathon October 2025
700 University Ave, Toronto, ON, Canada | github.com/AlezHibali/Interactive-Interview-Practice-with-Higgs
- **Top 16 Finalist** and **Special Mention Prize**.
- Designed and implemented an interactive **AI-powered interview coaching web application** providing real-time feedback on both technical content and speech delivery using LLMs and a RAG pipeline.
- Applied **advanced prompt engineering techniques**, including persona prompts, chain-of-thought guidance, structured JSON outputs, and few-shot examples, to produce consistent feedback and session summaries.
- Detailed explanations are available in the final report here.

Team Lead, AWS - VCT Hackathon: Esports Manager Challenge October 2024 – November 2024
Online | devpost.com/software/edg-learning
- **Top 25** Finalist out of 3,295 participants.
- Designed and implemented an **LLM-powered digital assistant** for building teams of professional players and answering questions about Valorant Esports.
- Leveraged large-scale esports data and **AWS Lambda and Bedrock** to tune the LLM agent through iterative prompting experiments and investigations.
- Deployed the application with Flask on AWS EC2 and automated **continuous deployment** using CodePipeline.
- Detailed explanations are available in the final report here.

TECHNICAL SKILLS

Programming: Python, C++, C, Java, Go, MATLAB, Shell **Web:** Flask, Django, Node.js, Vue.js, HTML, CSS
Generative AI: Large Language Models (LLMs), Diffusion Models, Transformers, RAG, Prompt Engineering, AI Agents
Computer Vision: Video Generation, 3D Vision, OpenCV, Unreal Engine 5 (UE5)
Deep Learning: PyTorch, TensorFlow, Diffusers, Hugging Face, Accelerate
Cloud & MLOps: AWS (EC2, Lambda, Bedrock, SageMaker, CodePipeline), Cloudflare R2, Docker